

Ethernet standard and specification

Task – 1 RJ connector labeling

- Pin Number	T568A Standard	T568B Standard (<i>Most Common</i>)
- Pin 1	White / Green	StripeWhite / Orange Stripe
- Pin 2	Solid Green	Solid Orange
- Pin 3	White / Orange	StripeWhite / Green Stripe
- Pin 4	Solid Blue	Solid Blue
- Pin 5	White / Blue Stripe	White / Blue Stripe
- Pin 6	Solid Orange	Solid Green
- Pin 7	White / Brown Stripe	White / Brown Stripe
- Pin 8	Solid Brown	Solid Brown



Task – 2 Specification of ethernet

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Specification	10BASE-T	100BASE-TX	1000BASE-T
Common Name	Standard Ethernet	Fast Ethernet	Gigabit Ethernet
Speed / Data Rate	10 Mbps	100 Mbps	1 Gbps (1,000 Mbps)

Minimum Cable Category	Category 3 (Cat3) or higher	Category 5 (Cat5) or higher	Category 5e (Cat5e) or higher
Pairs Used	2 Twisted Pairs	2 Twisted Pairs	4 Twisted Pairs
Max Cable Length	100 meters (328 feet)	100 meters (328 feet)	100 meters (328 feet)
Connector Type	8P8C / RJ-45	8P8C / RJ-45	8P8C / RJ-45

Task – 3 Connect the ethernet via cable

- To connect two laptops directly to one another using traditional Ethernet wiring, you use a Crossover Cable. Crossover Cable is used for network Interface Types (MDI vs. MDI-X): Laptop Network Interface Cards (NICs) are classified as MDI (Medium Dependent Interface) devices. These ports transmit data on Pins 1 & 2 and receive data on Pins 3 & 6.
- Laptop A's Transmit pins (1 & 2) connect directly to Laptop B's Transmit pins (1 & 2).
- Laptop A's Receive pins (3 & 6) connect directly to Laptop B's Receive pins (3 & 6).
- As a result, both devices speak into each other's "mouths" and listen to each other's "ears," blocking communication. The Crossover Solution: A crossover cable intentionally swaps the pairs on one end of the cable (T568A on one side, T568B on the other): Pins 1 & 2 (Transmit) on Laptop A connect to Pins 3 & 6 (Receive) on Laptop B. Pins 3 & 6 (Receive) on Laptop A connect to Pins 1 & 2 (Transmit) on Laptop B.
- Virtually all modern laptops and Gigabit Ethernet adapters (1000BASE-T) feature a technology called Auto MDI-X. If either laptop supports Auto MDI-X, it will automatically detect the direct connection and electronically swap the transmit and receive pairs internally, allowing you to use a standard Straight-Through cable as well.

Task – 4 Ethernet connections

Device	Likely Ethernet Standard	Data Rate / Cable Requirement
Desktop PC or Motherboard	1000BASE-T (<i>Gigabit Ethernet</i>) or 2.5GBASE-T	1 Gbps or 2.5 Gbps over Cat 5e/Cat 6 twisted pair cabling. Most modern consumer motherboards and standalone NICs support standard Gigabit (1000BASE-T), with newer enthusiast boards featuring 2.5GBASE-T.
Network-Attached Storage (NAS) or Campus Server	10GBASE-T or 1000BASE-T	10 Gbps (or 1 Gbps) over Cat 6A cabling. Used to handle simultaneous multi-user file transfers, data backups, and high-bandwidth media streaming without bottlenecking local network bandwidth.
Smart TV, IP Camera, or VoIP Desk Phone	100BASE-TX (<i>Fast Ethernet</i>)	100 Mbps over Cat 5 or Cat 5e cabling using 2 twisted pairs. Smart TVs, streaming sticks, and office IP phones often capped hardware at Fast Ethernet speeds because 100 Mbps exceeds the bitrates required for 4K video streaming or voice traffic.